When Are Sufficient Statistics Sufficient? Bounding the Approximation Error in Gains-from-Trade Formulas *

Yonggeun Jung†

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Abstract

The welfare formula of Arkolakis et al. (2012) is widely used to evaluate the gains from trade, but its validity rests on the assumption that productivity distributions are exactly Fréchet or Pareto. Empirical distributions systematically violate this requirement, introducing an approximation error of unknown magnitude. We derive strict upper bounds on this error for both the Eaton–Kortum and Melitz model classes, the two workhorses of modern quantitative trade analysis. The bounds depend solely on the variation of the trade elasticity across trade cost levels, an object directly estimable from standard gravity regressions without firm-level data, providing researchers with a practical tool to assess the reliability of their welfare calculations.

JEL Codes: F10, F14, C52

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*Replication package is available at https://github.com/yonggeun-jung/ACR_2012_approximation_public. The package includes all code and processed master files sufficient to reproduce all tables and figures. Raw data (CEPII Gravity database and BACI) must be downloaded separately from http://www.cepii.fr (publicly available). All remaining errors are our own. First version: March, 2026.

†Department of Economics, University of Florida. Email: yonggeun.jung@ufl.edu.

1 Introduction

Arkolakis et al. (2012) (hereafter ACR) establish one of the most influential results in modern trade theory: a broad class of models deliver the welfare formula:

$$\hat{W} = \hat{\lambda}^{-\frac{1}{\varepsilon}},$$

where $\hat{W}$ is the change in real income, $\hat{\lambda}$ is the change in the domestic expenditure share, and ε is the trade elasticity. The elegance and generality of this formula have made it a centerpiece of quantitative trade analysis. It serves as the primary welfare benchmark in evaluations of trade liberalisation (e.g., Costinot and Rodríguez-Clare, 2014), counterfactual analyses of regional trade agreements (Caliendo and Parro, 2015), and assessments of trade policy at the sectoral level (Ossa, 2015). Its appeal is that welfare calculations reduce to two observable objects, $\hat{\lambda}$ and ε, without requiring the full structure of the underlying model.

For the formula to be operational, however, ε must be a structural constant. In the Eaton–Kortum model (hereafter EK, Eaton and Kortum, 2002), this requires technology draws to follow a Fréchet distribution; in the Melitz model (Melitz, 2003), it requires firm-level productivity to follow a Pareto distribution. These distributional assumptions are not incidental modeling choices but mathematical necessities: without them, the trade elasticity varies along the equilibrium path, and the formula ceases to be exact.

That these assumptions are empirically tenuous is well understood. Evidence that productivity distributions depart from Fréchet and Pareto is extensive (e.g., Head et al., 2014; Fernandes et al., 2023). Melitz and Redding (2015) show that non-Pareto distributions generate new welfare margins absent from the ACR formula, and Arkolakis et al. (2019) explore how variable markups interact with distributional form. The qualitative message from this literature is clear: the ACR formula is an approximation, not an identity, and researchers who rely on it are implicitly betting that the distributional departure is small enough to be innocuous.

What is missing is a way to evaluate that bet. The existing literature identifies the problem but does not quantify it: given a particular dataset, how large could the ACR approximation error actually be? Without an answer, researchers face an uncomfortable choice between using a formula they know to be approximate and abandoning the tractability that makes it valuable in the first place.

This paper provides an upper bound on the ACR approximation error that researchers can compute alongside any welfare calculation. We maintain the four macroscopic ACR

conditions and allow the underlying distribution to be arbitrary. Because the distribution is no longer Fréchet or Pareto, the trade elasticity ε becomes endogenous and varies along the equilibrium path, and the ACR formula acquires an approximation error. We derive strict upper bounds on this error for both the EK and Melitz model classes. The bounds depend on a single macro-observable quantity: the variation of the gravity elasticity across trade cost levels. We validate the bounds through Monte Carlo simulations across four calibrated data-generating processes, and apply the diagnostic to the CEPII Gravity database at both the aggregate and HS2-sector level.

Our contributions to the literature are as follows. First, by recasting the micro-level bounds into macro-observable equivalents, we show that the distributional assumption underlying the ACR formula is testable without firm-level data. This lowers the barrier to robustness assessment: a researcher who reports ACR welfare gains can now report, at negligible additional cost, a quantitative bound on the error from the distributional assumption. Second, the local sufficiency conditions we derive (Corollaries 1 and 2) clarify when the Pareto or Fréchet assumption is innocuous and when it is not, providing theoretical guidance that complements the qualitative warnings in Head et al. (2014) and Melitz and Redding (2015) with quantitative content. Third, the sectoral application reveals that the reliability of the ACR formula varies sharply across industries: aggregate bilateral trade among the 30 largest economies is consistent with log-linear gravity, but roughly 40% of HS2 sectors exhibit statistically significant departures from constant elasticity, concentrated in agriculture and primary products. This finding offers concrete guidance to applied researchers on where the ACR formula can be trusted and where additional caution is warranted.

The remainder of this paper is organized as follows. Section 2 introduces the framework and defines the approximation error. Section 3 presents the theoretical bounds and their macro-observable reformulations. Section 4 describes the gravity-based diagnostic. Section 5 validates the bounds through Monte Carlo simulations. Section 6 provides the empirical application. Section 7 concludes.

2 Framework

We consider EK-class and Melitz-class trade models satisfying the four ACR conditions:

(i) CES preferences with elasticity of substitution $\sigma > 1$,

(ii) linear cost structure with iceberg trade costs $\tau_{ij} \geq 1$,

(iii) balanced trade, and

(iv) constant profit share.

Let $\lambda \in (0, 1)$ denote the domestic expenditure share. Under these conditions, the exact welfare change satisfies:

$$\ln \hat{W} = -\int_{\ln \lambda}^{\ln \lambda'} \frac{1}{\varepsilon(x)}\, dx, \tag{1}$$

where $\hat{W} \equiv W'/W$ is the ratio of post- to pre-shock real income and $\varepsilon(x) > 0$ is the trade elasticity at domestic share e^x. When ε is constant at $\varepsilon_{\text{base}}$, Equation (1) reduces to the ACR formula:

$$\ln \hat{W}_{\text{ACR}} = -\left(\frac{1}{\varepsilon_{\text{base}}}\right) \ln \hat{\lambda},$$

where $\hat{\lambda} \equiv \lambda'/\lambda$. The approximation error is:

$$\mathcal{E} \equiv \ln \hat{W}_{\text{exact}} - \ln \hat{W}_{\text{ACR}} = \int_{\ln \lambda}^{\ln \lambda'} \frac{\varepsilon(x) - \varepsilon_{\text{base}}}{\varepsilon_{\text{base}} \cdot \varepsilon(x)}\, dx, \tag{2}$$

measured in log points of real income.

2.1 EK Class

The trade elasticity depends on the technology distribution $F_0(z)$ with density $f_0(z)$, where $z > 0$ is a firm's efficiency draw. Under Fréchet ($F_0(z) = \exp(-z^{-\theta})$, shape $\theta > 0$), $\varepsilon = \theta$ is constant. Under a general F_0, the elasticity at margin z is the scaled hazard rate:

$$\varepsilon(z) = \frac{z\, f_0(z)}{1 - F_0(z)}, \tag{3}$$

and the hazard-rate deviation from Fréchet is $\eta(z) \equiv \varepsilon(z) - \theta$. The derivation connecting Equation (3) to the structure of the EK model is provided in Section A.1.

2.2 Melitz Class

Firms draw productivity $\varphi > 0$ from $G(\varphi)$ with density $g(\varphi)$. The trade elasticity at export cutoff φ^* is:

$$\varepsilon(\varphi^*) = (\sigma - 1) + \frac{\varphi^* g(\varphi^*)}{\Phi(\varphi^*)}, \tag{4}$$

where $\Phi(\varphi^*) \equiv \int_{\varphi^*}^{\infty} (\varphi/\varphi^*)^{\sigma-1} g(\varphi)\, d\varphi$ is the revenue-weighted productivity aggregate. Under Pareto ($\bar{G}(\varphi) = (\varphi_{\min}/\varphi)^{\kappa}$, $\kappa > \sigma - 1$), $\varepsilon = \kappa$ is constant. Under a general G, the elasticity varies with φ^*. The derivation is provided in Section A.2.

Melitz and Redding (2015) note that non-Pareto distributions introduce welfare margins through the endogenous mass of entrants. Our framework addresses this concern as follows. The four ACR conditions are maintained assumptions, not derived results. We do not claim that non-Pareto distributions generically satisfy the constant profit share condition. Rather, we ask: conditional on the four ACR conditions holding, what additional welfare error arises from the distributional departure? This is the relevant question for a researcher who has already adopted an ACR-class model (thereby imposing the four conditions) and wants to assess the sensitivity of the welfare calculation to the distributional assumption within that maintained framework. Under condition (iv), the equilibrium adjustment of the entry mass M_e is structurally absorbed into $\Phi(\varphi^*)$, so within this restricted class the variation in $\varepsilon(\varphi^*)$ alone captures the welfare approximation error.[1]

3 Theoretical Bounds

The approximation error $\mathcal{E}$ defined in Equation (2) depends on how much the trade elasticity ε deviates from its benchmark value along the equilibrium path. In this section, we formalize this dependence. We first derive bounds in terms of distributional primitives (Section 3.1), then reformulate them in terms of the observable trade elasticity alone (Section 3.2), and finally characterize the rate at which distributions converge to sufficiency (Section 3.3).

[1] More precisely, the constant profit share condition constrains the class of admissible non-Pareto distributions to those for which the entry margin does not independently affect welfare. This excludes, for example, distributions that generate a variable profit share through endogenous markup dispersion. The bounds we derive are therefore conditional on the maintained ACR restrictions, and should be interpreted as quantifying one specific source of approximation error (distributional departure) while holding other potential sources of error (profit share variation, markup dispersion) fixed.

3.1 Micro-Level Bounds

EK bound. In the EK class, the trade elasticity $\varepsilon(z) = \theta + \eta(z)$ varies with the marginal technology draw z. The question is: how much does this variation inflate the welfare error? Let $[z_L, z_H]$ denote the range of marginal draws traversed along the equilibrium path induced by the policy shock. We require a mild regularity condition:

Assumption 1. *$\varepsilon(z) \geq \sigma - 1$ for all $z \in [z_L, z_H]$, and $\theta \geq \sigma - 1$.*

Under Fréchet, the price index converges if and only if $\theta > \sigma - 1$ (Eaton and Kortum, 2002). EK extend this regularity condition to general distributions: it requires the trade elasticity to remain at least $\sigma - 1$ along the equilibrium path. This is economically natural. If $\varepsilon(z) < \sigma - 1$ at some margin, trade at that margin is less responsive to trade costs than the intensive margin alone ($\sigma - 1$), which is inconsistent with the comparative advantage mechanism that drives trade in the EK model. The condition is satisfied by all distributions in the domain of attraction of Fréchet with $\theta > \sigma - 1$, including all four DGPs considered in Section 5.

Proposition 1 (EK Bound)**.** *Under the four ACR conditions, Assumption 1, and Fréchet (shape θ) as benchmark:*

$$|\mathcal{E}_{\text{EK}}| \leq \sup_{z \in [z_L, z_H]} \frac{|\eta(z)|}{\theta + \eta(z)} \cdot |\ln \hat{\lambda}|. \tag{5}$$

See Section A.3 for the proof, which applies the triangle inequality to Equation (2) directly.

The bound has a transparent structure. The first factor, $\sup |\eta|/(\theta + \eta)$, measures the worst-case proportional deviation of the hazard rate from Fréchet along the equilibrium path. It equals zero under Fréchet and increases with the magnitude of the hazard-rate departure. The second factor, $|\ln \hat{\lambda}|$, scales the error by the size of the policy shock: larger shocks traverse a wider segment of the equilibrium path, exposing the welfare calculation to a greater range of elasticity variation.

Corollary 1 (EK Local Sufficiency)**.** *If $\eta(z) = 0$ for all $z \in [z_L, z_H]$, then $\mathcal{E}_{\text{EK}} = 0$ regardless of the global shape of F_0.*

Corollary 1 states that the ACR formula can be locally exact even if the technology distribution departs from Fréchet globally. What matters is whether the departure occurs in the region of the distribution that is actually affected by the policy shock. A distribution that is non-Fréchet in the left tail but Fréchet-like in the right tail will generate zero ACR error for shocks that only move cutoffs within the right tail.

Melitz bound. The Melitz class introduces a richer structure because the trade elasticity in Equation (4) depends on the ratio of two objects: the local density $g(\varphi^*)$ and the revenue-weighted tail aggregate $\Phi(\varphi^*)$. To isolate the sources of departure, we decompose each relative to its Pareto counterpart. Let $g_P(\varphi) = \kappa\varphi_{\min}^{\kappa}\varphi^{-(\kappa+1)}$ and $\Phi_P(\varphi^*) = \kappa(\kappa - \sigma + 1)^{-1}(\varphi_{\min}/\varphi^*)^{\kappa}$ denote the Pareto density and revenue aggregate.

Definition 1 (Density and Tail Deviations)**.** $\rho(\varphi^*) \equiv g(\varphi^*)/g_P(\varphi^*) - 1$ *and* $\psi(\varphi^*) \equiv \Phi(\varphi^*)/\Phi_P(\varphi^*) - 1$.

Economically, ρ captures how much the mass of marginal firms (those at the cutoff) deviates from Pareto, while ψ captures how much the revenue contribution of all inframarginal firms (those above the cutoff) deviates. Substituting into Equation (4), the elasticity departure from Pareto is:

$$\varepsilon(\varphi^*) - \kappa = [\kappa - (\sigma - 1)] \cdot \frac{\rho(\varphi^*) - \psi(\varphi^*)}{1 + \psi(\varphi^*)}. \tag{6}$$

Equation (6) reveals that the elasticity departure is governed by the wedge $\rho - \psi$, not by ρ or ψ individually. A distribution with excess mass at the cutoff ($\rho > 0$) but a proportionally heavier tail ($\psi = \rho$) generates no elasticity departure at all. This is economically natural: what distorts the welfare calculation is not whether there are "too many" or "too few" firms relative to Pareto, but whether the density and the revenue-weighted tail are distorted asymmetrically.

Assumption 2. $\Phi(\varphi) > 0$ *for all* $\varphi \in [\varphi_L, \varphi_H]$.

Assumption 2 requires a positive mass of firms above the cutoff throughout the equilibrium path. It is violated only if the distribution has finite support and the cutoff reaches the upper bound during the transition.

Proposition 2 (Melitz Bound)**.** *Under the four ACR conditions, Assumption 2, and* $\kappa > \sigma - 1$*:*

$$|\mathcal{E}_{\text{Melitz}}| \leq \frac{\kappa - (\sigma - 1)}{(\sigma - 1)\,\kappa} \cdot \sup_{\varphi \in [\varphi_L, \varphi_H]} \frac{|\rho(\varphi) - \psi(\varphi)|}{1 + \psi(\varphi)} \cdot |\ln \hat{\lambda}|. \tag{7}$$

See Section A.4 for the proof.

Comparing Equations (5) and (7) highlights an important asymmetry between the two model classes. Under Assumption 1, both bounds share the structural floor at $\sigma - 1$, but they differ in their sensitivity to distributional departures. The EK diagnostic bound (Equation (16)) scales as $\widehat{\Delta\varepsilon}/(\sigma - 1)$, while the Melitz bound (Equation (17)) scales as

$\widehat{\Delta\varepsilon}/[(\sigma-1)\hat{\varepsilon}_{\text{pool}}]$. Since $\hat{\varepsilon}_{\text{pool}} > 1$ in practice, the Melitz bound is tighter by a factor of $\hat{\varepsilon}_{\text{pool}}$. This reflects the additional dampening from the revenue-weighted aggregate Φ in the Melitz elasticity: the $(\sigma-1)$ floor in Equation (4) compresses the elasticity variation relative to the EK class, where the hazard rate enters without dampening.

Corollary 2 (Melitz Local Sufficiency)**.** *If $\rho(\varphi) = \psi(\varphi)$ for all $\varphi \in [\varphi_L, \varphi_H]$, then $\mathcal{E}_{\text{Melitz}} = 0$.*

As with Corollary 1, the condition is local: the density and tail deviations need only coincide along the segment of the distribution traversed by the policy shock.

3.2 Macro-Observable Reformulation

The micro-level bounds require firm-level distributional data to estimate η, ρ, and ψ. This limits their practical applicability, since the researchers who use the ACR formula typically lack access to firm-level productivity data. We now show that the bounds admit equivalent reformulations in terms of ε alone. This is the key theoretical step: it connects the distributional primitives to an object that is directly estimable from aggregate bilateral trade data, and thereby enables the gravity-based diagnostic developed in Section 4.

Proposition 3 (Macro EK Bound)**.** *Under the conditions of Proposition 1:*

$$|\mathcal{E}_{\text{EK}}| \leq \sup_{z\in[z_L,z_H]} \frac{|\varepsilon(z)-\theta|}{\varepsilon(z)} \cdot |\ln \hat{\lambda}|. \tag{8}$$

Proof. Substitute $\eta(z) = \varepsilon(z) - \theta$ into Equation (5). □

Corollary 3 (Tightened EK Bound)**.** *Under Assumption 1 with $\varepsilon(z) \geq \sigma - 1$:*

$$|\mathcal{E}_{\text{EK}}| \leq \frac{\sup_z |\varepsilon(z)-\theta|}{\sigma-1} \cdot |\ln \hat{\lambda}|. \tag{9}$$

Proof. From Equation (8), replace $\varepsilon(z)$ in the denominator with its lower bound $\sigma-1$. □

Corollary 3 brings the EK bound into the same scale as the Melitz bound (Equation (10)), with both denominators proportional to $\sigma-1$. The key difference is that the Melitz bound carries the additional factor $1/\kappa$, reflecting the dampening from the revenue-weighted aggregate Φ. When Assumption 1 is violated, Corollary 3 does not apply, but the original bound in Proposition 3 remains valid with $\varepsilon(z)$ in the denominator.

Proposition 4 (Macro Melitz Bound). *Under the conditions of Proposition 2:*

$$|\mathcal{E}_{\text{Melitz}}| \leq \frac{1}{(\sigma-1)\,\kappa} \cdot \sup_{\varphi^*} |\varepsilon(\varphi^*) - \kappa| \cdot |\ln \hat{\lambda}|. \tag{10}$$

Proof. Substitute Equation (6) into Equation (7). □

The practical import of Equations (8) and (10) is that neither η, ρ, nor ψ appears. The bounds depend entirely on the variation of the aggregate trade elasticity ε relative to its benchmark value. Under Fréchet or Pareto, ε is constant, so any observed variation in ε across trade cost levels is diagnostic of distributional departure, and the magnitude of that variation is all a researcher needs to bound the welfare error.

3.3 Rate of Convergence to Sufficiency

The sufficiency conditions ($\eta = 0$ in EK, $\rho = \psi$ in Melitz) may appear restrictive. In fact, they are not knife-edge: many commonly used distributions satisfy them approximately in the upper tail, and the approximation improves as the cutoff moves deeper into the tail. We formalize this using second-order regular variation (De Haan and Ferreira, 2006), which provides a natural framework for quantifying the rate at which a distribution converges to its Fréchet/Pareto limit.

Definition 2 (Second-Order Regular Variation). $\bar{G} \in 2\text{RV}(-\kappa, \beta)$ *with* $\beta \leq 0$ *if there exists* $A(\varphi) \to 0$, $|A| \in \text{RV}(\beta)$, *such that for all* $t > 0$*:*

$$\lim_{\varphi \to \infty} \frac{\bar{G}(t\varphi)/\bar{G}(\varphi) - t^{-\kappa}}{A(\varphi)} = t^{-\kappa} \cdot \frac{t^{\beta} - 1}{\beta}. \tag{11}$$

The first-order parameter κ *is the Pareto tail index. The second-order parameter* $\beta \leq 0$ *governs the convergence rate:* β *closer to zero means slower convergence. The auxiliary function* $A(\varphi) \to 0$ *captures the local magnitude of the departure from exact Pareto behavior.*

Lemma 1 (Convergence Rate). *If the technology distribution satisfies* $F_0 \in 2\text{RV}(-\theta, \beta)$ *with auxiliary function* A*, then* $|\eta(z)|/(\theta + \eta(z)) = O(|A(z)|)$ *as* $z \to \infty$*. Analogously, if* $\bar{G} \in 2\text{RV}(-\kappa, \beta)$*, then* $\rho(\varphi^*) - \psi(\varphi^*) = O(A(\varphi^*))$ *as* $\varphi^* \to \infty$*.*

See Section A.5 for the proof.

Lemma 1 has two practical implications. First, for distributions with slow tail convergence, the ACR error may be non-negligible even at high cutoffs. The leading example is the log-normal distribution, which has $\beta = 0$ and $|A(\varphi)| = O(1/\ln \varphi)$: the

sufficiency wedge $|\rho - \psi|$ shrinks only logarithmically, so it persists at empirically relevant cutoffs. This is consistent with the well-documented difficulty of distinguishing log-normal from Pareto tails in firm-size data (Head et al., 2014). Second, for distributions with rapid convergence, the ACR formula becomes reliable even if the distribution is globally non-Pareto. The truncated Pareto ($\beta < 0$, $|A(\varphi)| \sim \varphi^{\beta}$) is the canonical example: the wedge shrinks polynomially, and the ACR error vanishes rapidly as the cutoff increases.

4 Gravity-Based Diagnostic

The macro-observable bounds in Equations (8) and (10) require the variation of ε across trade cost levels. We now describe a concrete procedure to estimate this variation from bilateral trade data and evaluate the bound. The structural derivations connecting the trade elasticity formulas in Equations (3) and (4) to gravity are provided in Sections A.1 and A.2; here we focus on the empirical implementation.

From trade costs to gravity. In both model classes, bilateral trade depends on iceberg trade costs τ_{ij}, and the gravity elasticity $-\partial \ln X_{ij}/\partial \ln \tau_{ij} = \varepsilon$ is constant under Fréchet/Pareto. In empirical gravity equations, bilateral distance d_{ij} proxies for trade costs: $\ln \tau_{ij} = \delta \ln d_{ij} + \ln h_{ij}$ for some $\delta > 0$, where h_{ij} collects non-distance barriers absorbed by fixed effects. The standard gravity regression:

$$\ln X_{ij} = S_i + M_j - \varepsilon \ln d_{ij} + u_{ij} \tag{12}$$

(with exporter and importer fixed effects S_i, M_j, respectively) recovers a constant distance coefficient under Fréchet/Pareto. Under non-standard distributions, the gravity elasticity varies with d_{ij}, inducing a nonlinear relationship between log trade and log distance.

Procedure. The diagnostic consists of three steps.[2]

Step 1: Pooled gravity. Estimate Equation (12) by Poisson Pseudo Maximum Likelihood (PPML) to obtain the benchmark elasticity $\hat{\varepsilon}_{\text{pool}} \equiv -\hat{\beta}_1$ (Silva and Tenreyro, 2006). This is

[2] An alternative to Step 2 is to partition bilateral pairs into trade cost quantiles and estimate separate elasticities within each bin. We adopt the quadratic specification as the baseline because it uses the full sample efficiently. Any method that reliably identifies elasticity variation can serve the same purpose.

the constant elasticity that enters the standard ACR formula and serves as the baseline against which departures are measured.

Step 2: Augmented gravity. Re-estimate the gravity equation with a quadratic distance term:

$$X_{ij} = \exp\big(S_i + M_j + \beta_1 \ln d_{ij} + \beta_2 (\ln d_{ij})^2\big) \cdot u_{ij}. \tag{13}$$

Under Fréchet/Pareto, $\beta_2 = 0$ and the gravity relationship is exactly log-linear. If $\hat{\beta}_2$ is statistically different from zero, the trade elasticity varies across bilateral pairs:

$$\varepsilon(d_{ij}) = -(\beta_1 + 2\beta_2 \ln d_{ij}). \tag{14}$$

The implied range of elasticity variation across observed distances is:

$$\widehat{\Delta\varepsilon} = 2|\hat{\beta}_2| \cdot (\ln d_{\max} - \ln d_{\min}). \tag{15}$$

The test $H_0\colon \beta_2 = 0$ is thus a direct test of the Fréchet/Pareto distributional assumption, conditional on the four ACR conditions.

Step 3: Bound evaluation. Substitute $\widehat{\Delta\varepsilon}$ into the macro-observable bounds. For EK, applying Corollary 3:

$$|\hat{\mathcal{E}}_{\mathrm{EK}}| \leq \frac{\widehat{\Delta\varepsilon}}{\sigma - 1} \cdot |\ln \hat{\lambda}|. \tag{16}$$

For Melitz:

$$|\hat{\mathcal{E}}_{\mathrm{Melitz}}| \leq \frac{\widehat{\Delta\varepsilon}}{(\sigma - 1)\,\hat{\varepsilon}_{\mathrm{pool}}} \cdot |\ln \hat{\lambda}|. \tag{17}$$

All inputs are available from Steps 1 and 2 ($\hat{\varepsilon}_{\mathrm{pool}}$, $\hat{\beta}_2$, distance range), the elasticity of substitution σ, and the observed change in the domestic expenditure share $\hat{\lambda}$. If $\hat{\beta}_2$ is insignificant, the researcher concludes that the ACR formula is reliable at the precision afforded by the data. If $\hat{\beta}_2$ is significant, the bound quantifies the worst-case welfare error and the researcher can assess whether it is economically tolerable for the application at hand.

Cross-sectional variation as a bound on the counterfactual path. The theoretical error in Equation (2) is defined along the counterfactual equilibrium path $[\ln \lambda, \ln \lambda']$, whereas the diagnostic estimates elasticity variation across bilateral pairs in the cross section. The connection between these two objects rests on the observation that, in the ACR class, different bilateral pairs face different trade costs and hence different marginal cutoffs φ^*_{ij}. The cross-sectional variation in $\varepsilon(d_{ij})$ therefore traces out the same function $\varepsilon(\cdot)$ that governs the counterfactual path, evaluated at different points of its domain. The diagnostic bound is valid as long as the counterfactual policy shock does not move equilibrium cutoffs outside the range spanned by observed bilateral distances. For shocks that alter trade costs within the range $[d_{\min}, d_{\max}]$ of existing bilateral pairs, the cross-sectional $\widehat{\Delta\varepsilon}$ provides a valid upper bound on the path-level variation. For shocks that create entirely new trading relationships at distances outside this range, the bound may understate the true error, and extrapolation of the estimated $\varepsilon(d)$ profile should be treated with caution.

Identification. A natural question is whether the test $H_0\colon \beta_2 = 0$ has a structural interpretation beyond a statistical specification check. The following result confirms that, within the ACR class, gravity curvature is equivalent to distributional departure.

Remark 1 (Identification)**.** *Under the four ACR conditions and a continuously differentiable distribution, $\beta_2 = 0$ in Equation (13) if and only if the distribution is Fréchet (EK) or Pareto (Melitz) on the support spanned by observed bilateral pairs. The argument is straightforward: in the EK case, $\beta_2 = 0$ implies $\varepsilon(z) = \theta$ for all observed margins, which from Equation (3) yields $d\ln(1 - F_0)/d\ln z = -\theta$ with unique solution $F_0(z) = \exp(-z^{-\theta})$. The Melitz case is analogous (see Section A.6).*

If the ACR conditions themselves are violated, $\beta_2 \neq 0$ still implies that the ACR formula with constant ε is unreliable, and the computed bound provides a lower bound on the total welfare error.

Limitations. Three caveats apply. First, the theoretical bounds in Propositions 1 and 2 involve a supremum over the full equilibrium path, whereas the empirical diagnostic estimates $\widehat{\Delta\varepsilon}$ from the observed distance range $[\ln d_{\min}, \ln d_{\max}]$. If the true distribution exhibits larger departures at cutoff levels not spanned by observed bilateral pairs (e.g., in the extreme tails), the empirical bound may understate the theoretical supremum. This limitation is inherent in any finite-sample diagnostic and is mitigated by the wide distance range in our sample (174 km to 19,812 km), which spans nearly the full range of feasible

bilateral distances. Second, the quadratic specification imposes a smooth parametric form on $\varepsilon(d)$ and may fail to detect localized departures (e.g., density bunching at specific productivity thresholds). The quantile-based alternative mentioned in footnote 2 provides a nonparametric check at the cost of reduced precision. Third, if the observed curvature partly reflects non-distributional sources of trade cost nonlinearity (e.g., non-iceberg frictions or multilateral resistance misspecification), the distributional component of $\widehat{\Delta\varepsilon}$ is smaller than the total, and the reported bound overstates the true ACR error. In this sense, alternative explanations for gravity curvature make our diagnostic more conservative, not less.

5 Monte Carlo Validation

We simulate symmetric 30-country economies under four data-generating processes (DGPs), each representing a distinct pattern of distributional departure from the Fréchet/Pareto benchmark. In all cases we set $\sigma = 4$ and draw bilateral trade costs uniformly from $[1.1, 3.0]$. Since the EK and Melitz models generate different trade flow structures from the same underlying distribution, we run the diagnostic separately for each model class: EK trade flows are proportional to the survival function $1 - F_0(z)$, while Melitz trade flows are proportional to the revenue-weighted aggregate $\Phi(\varphi^*)$. The four DGPs are:

(i) *Fréchet/Pareto* ($\theta = \kappa = 5$). The exact benchmark: the trade elasticity is constant at 5, and $\hat{\beta}_2 = 0$ by construction. Any nonzero error or bound in this case reflects only numerical precision.

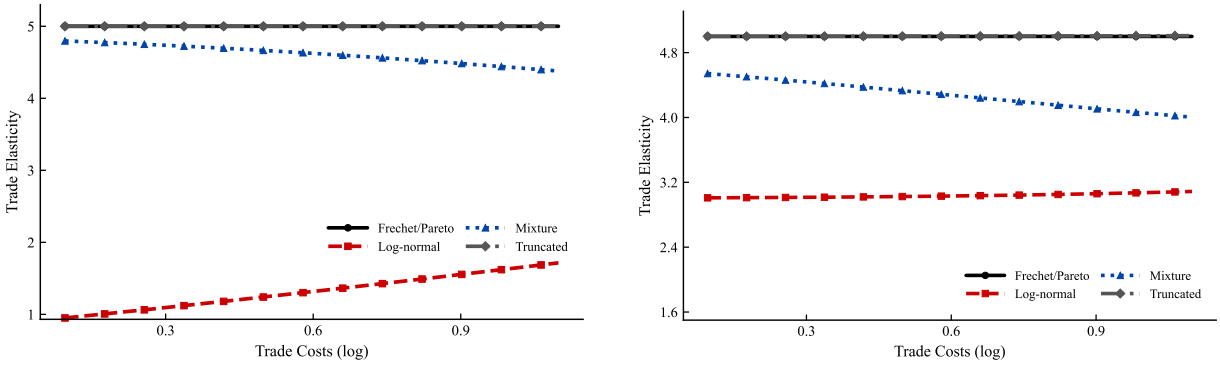
(ii) *Log-normal* ($\mu = 0$, $\sigma_{\ln} = 1$). A distribution with no power-law tail. The hazard rate varies continuously across the support, so the EK elasticity departs substantially from any constant. The Melitz elasticity is dampened by the $\sigma - 1$ floor but still varies. This DGP represents the empirically relevant scenario in which the Pareto approximation is poor throughout the distribution.

(iii) *Pareto mixture* ($\kappa_1 = 5$, $\kappa_2 = 3.5$, $\pi = 0.9$). A 90–10 mixture of two Pareto distributions with different shape parameters. The heavier-tailed component ($\kappa_2 = 3.5$) contaminates the benchmark ($\kappa_1 = 5$), creating localized density bunching that the diagnostic should detect. This DGP captures scenarios where a subpopulation of firms (e.g., multinationals) follows a different productivity distribution.

(iv) *Truncated Pareto* ($\kappa = 5$, $\bar{\varphi} = 50$). A Pareto distribution with an upper bound on productivity. The truncation introduces a second-order departure that shrinks polynomially in the cutoff (Lemma 1), so the ACR formula should remain accurate. This DGP tests whether the diagnostic correctly identifies benign departures.

For each DGP, we compute population-level trade flows and apply the three-step gravity diagnostic of Section 4.

Figure 1 plots the gravity elasticity profile for each DGP. Under Fréchet/Pareto, ε is flat in both panels. The key difference between the two model classes is visible under the log-normal DGP: in the EK class (panel a), the elasticity drops sharply across the trade cost range, whereas in the Melitz class (panel b) the variation is compressed by the structural floor at $\sigma - 1$. Under the mixture, ε varies substantially in both classes. The truncated Pareto is indistinguishable from the benchmark.



(a) EK class

(b) Melitz class

Figure 1: Trade Elasticity Profiles across DGPs

Table 1 reports the curvature coefficient $\hat{\beta}_2$ and implied elasticity variation for both model classes. Under Fréchet/Pareto, $\hat{\beta}_2 = 0$ exactly in both cases. The most striking difference appears under the log-normal DGP: the EK diagnostic detects substantial curvature ($\hat{\beta}_2 = -0.384$, $\widehat{\Delta\varepsilon} = 0.768$), whereas the Melitz diagnostic finds only modest curvature ($\hat{\beta}_2 = -0.038$, $\widehat{\Delta\varepsilon} = 0.077$). This reflects the structural floor at $\sigma - 1$ in the Melitz elasticity (Equation (4)), which compresses the range of elasticity variation relative to the EK class.

Table 2 compares true ACR errors to the diagnostic bounds for both model classes. All bounds hold across all DGPs and shock sizes. Two patterns are noteworthy. First, the EK class is far more vulnerable to distributional departures than the Melitz class. Under the log-normal DGP, the EK error reaches 13.6% of real income for a 40% trade cost reduction, compared to just 0.21% in the Melitz class. Although Assumption 1

Table 1: Monte Carlo: gravity diagnostic

	EK class			Melitz class		
DGP	$\hat{\varepsilon}_{\text{pool}}$	$\hat{\beta}_2$	$\widehat{\Delta\varepsilon}$	$\hat{\varepsilon}_{\text{pool}}$	$\hat{\beta}_2$	$\widehat{\Delta\varepsilon}$
Fréchet/Pareto	5.000	0.000	0.000	5.000	0.000	0.000
Log-normal	1.340	−0.384	0.768	3.036	−0.038	0.077
Mixture ($\pi = 0.9$)	4.606	0.213	0.426	4.260	0.276	0.552
Truncated ($\bar{\varphi} = 50$)	5.000	0.000	0.000	5.004	−0.004	0.008

Notes: Population-level gravity diagnostic on simulated trade data. EK trade flows use the survival function $1 - F_0(z)$; Melitz trade flows use the revenue-weighted aggregate $\Phi(\varphi^*)$. $\hat{\beta}_2 = 0$ under Fréchet/Pareto in both classes. Parameters: $\sigma = 4$, $\theta = \kappa = 5$, $N = 30$, $\tau \in [1.1, 3.0]$.

imposes a structural floor at $\sigma - 1$ on both model classes, the EK bound (Equation (16)) scales as $\widehat{\Delta\varepsilon}/(\sigma - 1)$, whereas the Melitz bound (Equation (17)) carries the additional dampening factor $1/\hat{\varepsilon}_{\text{pool}}$. Under the log-normal DGP, the EK diagnostic detects much larger elasticity variation ($\widehat{\Delta\varepsilon} = 0.768$) than the Melitz diagnostic ($\widehat{\Delta\varepsilon} = 0.077$), yielding a bound ratio of roughly 30:1. Note that the log-normal DGP violates Assumption 1 in the EK class ($\varepsilon_{\min} = 0.94 < \sigma - 1 = 3$), so the EK bound for this DGP is computed from Proposition 3 rather than Corollary 3; it nonetheless holds in all cases.

Second, the bounds are conservative but informative. Under the mixture DGP, the Melitz bound at a 40% reduction is 7.58% versus a true error of 1.07%, a ratio of roughly 7:1. This conservatism is inherent in any supremum-based bound, but the bound nonetheless correctly identifies the mixture as the most problematic DGP and provides a quantitative ceiling that a researcher can evaluate against the magnitude of the welfare calculation.

6 Empirical Application

We apply the diagnostic to the CEPII Gravity database (Conte et al., 2022) at both the aggregate and sectoral level, using bilateral trade flows (from BACI), geodesic distances, and GDP for the 30 largest economies by average GDP over 2015–2019, yielding 870 bilateral pairs per year (30×29). We construct domestic expenditure shares as $\lambda_{it} = 1 - \sum_{j \neq i} X_{ijt}/\text{GDP}_{it}$; the mean λ is approximately 0.78, and the median year-to-year change is $|\Delta \ln \lambda| = 0.013$.

Table 2: Monte Carlo: ACR welfare error and diagnostic bounds

DGP		EK class			Melitz class		
		10%	25%	40%	10%	25%	40%
Fréchet/Pareto	Error	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%
	Bound	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%
Log-normal	Error	0.11%	2.85%	13.61%	0.01%	0.07%	0.21%
	Bound[a]	6.61%	18.83%	34.71%	0.22%	0.62%	1.15%
Mixture ($\pi = 0.9$)	Error	<0.01%	0.17%	0.70%	0.01%	0.23%	1.07%
	Bound	4.01%	12.50%	24.91%	1.22%	3.80%	7.58%
Truncated ($\bar{\varphi} = 50$)	Error	<0.01%	<0.01%	<0.01%	0.00%	0.00%	0.01%
	Bound	<0.01%	<0.01%	<0.01%	0.02%	0.05%	0.10%

Notes: Error is $|\ln \hat{W}_{\text{exact}} - \ln \hat{W}_{\text{ACR}}|$ as a percentage of real income. EK bound from Equation (16); Melitz bound from Equation (17). Each model class uses its own diagnostic ($\widehat{\Delta\varepsilon}$, $\hat{\varepsilon}_{\text{pool}}$) estimated from the corresponding trade flow structure. Trade cost reduced from 2.0 to 1.8/1.5/1.2. Parameters: $\sigma = 4$, $\theta = \kappa = 5$, $N = 30$. [a]For the log-normal EK bound, Assumption 1 is violated; the bound is computed from Proposition 3.

6.1 Aggregate Results

For each year 2015–2019, we estimate Equations (12) and (13) by PPML with exporter and importer fixed effects, clustering standard errors at the pair level.

Table 3 reports the results. The curvature coefficient $\hat{\beta}_2$ is statistically insignificant in all years ($p > 0.47$), consistent with log-linear gravity. The implied bounds are small: 0.03–0.06 pp (EK) and 0.04–0.08 pp (Melitz) at the data-implied $|\Delta \ln \hat{\lambda}| = 0.013$. The pooled elasticity $\hat{\varepsilon}_{\text{pool}} \approx 0.78$ is below conventional estimates ($\varepsilon \in [3, 8]$; see Head et al., 2014), reflecting the parsimonious specification without bilateral controls beyond distance. Since the diagnostic depends on curvature $\hat{\beta}_2$ rather than the level of $\hat{\varepsilon}_{\text{pool}}$, and a lower $\hat{\varepsilon}_{\text{pool}}$ inflates the denominator of Equation (17), the reported bounds are conservative. Results are robust to σ (see Table B1 in the Appendix).

6.2 Sectoral Results

The aggregate results may mask substantial heterogeneity across industries. To investigate, we apply the diagnostic at the HS 2-digit sectoral level using bilateral trade from BACI HS02 for 2019. For each of the 97 HS2 sectors, we estimate Equations (12) and (13) separately, restricting to sectors with at least 100 bilateral pairs.

Table 4 presents selected sectors that illustrate the cross-industry pattern. Out of 90

Table 3: Aggregate gravity diagnostic, 2015–2019

			Elasticity range		ACR bound (pp)	
Year	$\hat{\varepsilon}_{\text{pool}}$	$\hat{\beta}_2$ (p-value)	$\varepsilon_{\min}$	$\varepsilon_{\max}$	EK	Melitz
2015	0.788	−0.022 (0.52)	0.668	0.879	0.06	0.07
2016	0.778	−0.025 (0.47)	0.645	0.879	0.06	0.08
2017	0.784	−0.015 (0.67)	0.704	0.844	0.04	0.05
2018	0.782	−0.011 (0.76)	0.723	0.826	0.03	0.04
2019	0.769	−0.022 (0.53)	0.653	0.858	0.06	0.07

Notes: PPML with exporter and importer fixed effects, pair-clustered standard errors. $\hat{\beta}_2$: coefficient on $(\ln d_{ij})^2$; p-values test $H_0\colon \beta_2 = 0$. Bounds evaluated at $\sigma = 4$ and $|\Delta \ln \hat{\lambda}| = 0.013$ (median from data). Sample: 30 countries, 870 pairs/year.

sectors with sufficient observations, 39 (43%) reject the null $H_0\colon \beta_2 = 0$ at the 5% level. The full results are reported in Table B2 in the Appendix.

Two patterns stand out. First, the departures are concentrated in agriculture and primary products. Sectors such as fats and oils (HS 15), coffee and spices (HS 09), cereals (HS 10), and food residues (HS 23) all exhibit large and highly significant curvature coefficients ($p < 0.001$ in each case), with implied elasticity variation $\widehat{\Delta\varepsilon}$ exceeding 1.5. The EK bounds for these sectors are roughly 3–5 times larger than the Melitz bounds, reaching 0.4–0.6 pp compared to 0.11–0.13 pp. The difference reflects the additional dampening factor $1/\hat{\varepsilon}_{\text{pool}}$ in the Melitz bound (Equation (17)). Both bounds remain below 1 pp at the data-implied $|\Delta \ln \hat{\lambda}| = 0.013$, but for larger policy shocks the proportional difference between EK and Melitz bounds would persist.

Second, differentiated manufactures exhibit no significant departures. Electrical machinery (HS 85), glass (HS 70), and rubber (HS 40) all have $|\hat{\beta}_2| < 0.005$ with p-values exceeding 0.90. Both EK and Melitz bounds for these sectors are below 0.05 pp. For these sectors, the ACR formula is essentially exact at observed liberalisation magnitudes.

The economic logic behind this pattern is intuitive. In differentiated manufacturing, the combination of strong firm heterogeneity and intense selection into exporting naturally generates Pareto-like upper tails, consistent with the theoretical predictions of Helpman et al. (2004) and the empirical findings of Eaton et al. (2011). The Pareto approximation holds well precisely where it was designed to work: in industries with many heterogeneous firms competing on productivity. In agriculture and primary products, by contrast, trade patterns are shaped by factors that Pareto cannot capture: geographic concentration of production (fisheries near coastlines, oil seeds in specific

climatic zones), perishability that imposes natural distance cutoffs beyond the smooth iceberg cost structure, and non-tariff barriers such as sanitary and phytosanitary measures that create discontinuities in market access. These forces generate productivity distributions with thinner tails or localized mass concentrations, precisely the patterns that our diagnostic detects through gravity curvature.

The sign of $\hat{\beta}_2$ is itself informative. In sectors with $\hat{\beta}_2 < 0$ (e.g., dairy products (HS 04) and ores (HS 26); see Table B2), the gravity elasticity increases with distance: the trade penalty steepens for distant pairs, consistent with perishability or bulk transport costs selecting out all but the most productive exporters at long distances. In sectors with $\hat{\beta}_2 > 0$ (e.g., cereals (HS 10) and coffee (HS 09)), the elasticity decreases with distance: nearby pairs trade disproportionately more, consistent with geographic concentration of production that creates excess mass at moderate productivities relative to the Pareto tail.

Table 4: Sectoral gravity diagnostic (selected HS2 sectors, 2019)

					ACR bound (pp)	
HS2	Sector	$\hat{\varepsilon}_{\text{pool}}$	$\hat{\beta}_2$	$\widehat{\Delta\varepsilon}$	EK	Melitz
Significant departures						
15	Fats and oils	1.034	0.232^{***}	2.196	0.60	0.13
09	Coffee, tea, spices	0.915	0.174^{***}	1.644	0.45	0.12
23	Food residues	0.976	0.215^{***}	2.028	0.55	0.13
10	Cereals	1.280	0.210^{***}	1.985	0.54	0.12
44	Wood	0.958	0.177^{***}	1.675	0.46	0.12
16	Meat/fish prep.	1.152	0.157^{***}	1.488	0.41	0.11
No significant departures						
85	Electrical machinery	0.826	−0.001	0.010	<0.01	<0.01
70	Glass	0.994	−0.000	0.001	<0.01	<0.01
40	Rubber	0.857	−0.005	0.045	0.01	0.02
73	Iron/steel articles	0.991	0.031	0.293	0.08	0.05

Notes: PPML with exporter and importer fixed effects, pair-clustered standard errors. ***: $p < 0.01$. Bounds evaluated at $\sigma = 4$ and $|\Delta \ln \hat{\lambda}| = 0.013$. Out of 90 sectors with ≥ 100 observations, 39 (43%) reject $H_0\colon \beta_2 = 0$ at 5%. Full results in Table B2. Sample: 30 countries, 2019.

7 Conclusion

This paper derives strict upper bounds on the ACR welfare formula's approximation error when Fréchet or Pareto distributional assumptions are violated, while maintaining

the four ACR macro-restrictions. The bounds depend on a single macro-observable quantity: the variation of the trade elasticity across trade cost levels. This yields a diagnostic tool that researchers can implement using standard gravity data without firm-level microdata.

The empirical application reveals a nuanced picture. At the aggregate level, bilateral trade among major economies is consistent with log-linear gravity, and ACR welfare calculations are reliable with error bounds below 0.1 percentage points. At the sectoral level, however, nearly half of HS2 sectors exhibit statistically significant departures from constant elasticity. These departures are concentrated in agriculture and primary products, where production is geographically concentrated, goods are perishable, and non-tariff barriers create discontinuities in market access. For these sectors, the ACR formula with a constant trade elasticity may be misleading. Differentiated manufactures, by contrast, show no detectable departure, consistent with selection-driven Pareto tails in industries with heterogeneous firms.

As a practical guideline, we suggest that researchers compare the computed bound to the magnitude of the welfare effect being reported. If the bound is less than 10% of the estimated welfare gain, the ACR formula can be used with confidence; if it exceeds this threshold, the researcher should consider whether the distributional assumption is consequential for the policy conclusion at hand.

Several extensions merit investigation. First, extending the bounds to variable markup environments (Arkolakis et al., 2019) would decompose welfare errors into distributional and markup components. Second, time-series application of the diagnostic could track whether distributional departures grow or shrink with trade liberalisation episodes. Third, the sectoral patterns suggest a connection between product characteristics (homogeneity, perishability, regulatory barriers) and distributional form that deserves theoretical formalization.

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A Proofs and Derivations

A.1 Derivation of the EK Trade Elasticity

In the EK model, country j sources each good ω from the lowest-cost supplier. Country i offers good ω in market j at price

$$p_{ij}(\omega) = \frac{c_i\,\tau_{ij}}{z_i(\omega)}, \tag{A1}$$

where c_i is the unit input cost, $\tau_{ij} \geq 1$ is the iceberg trade cost, and $z_i(\omega)$ is an i.i.d. efficiency draw from F_0. Country j imports from the cheapest source. The share of goods for which i is the lowest-cost supplier determines bilateral trade, which satisfies

$$X_{ij} \propto 1 - F_0(\tilde{z}_{ij}), \tag{A2}$$

where $\tilde{z}_{ij}$ is the marginal technology draw for pair (i, j), with $\tilde{z}_{ij}$ increasing in τ_{ij}. The gravity elasticity is therefore

$$-\frac{\partial \ln X_{ij}}{\partial \ln \tau_{ij}} = \frac{\tilde{z}_{ij}\, f_0(\tilde{z}_{ij})}{1 - F_0(\tilde{z}_{ij})} = \varepsilon(\tilde{z}_{ij}), \tag{A3}$$

which is Equation (3). Under Fréchet, $F_0(z) = \exp(-z^{-\theta})$, so

$$f_0(z) = \theta\, z^{-(\theta+1)} \exp(-z^{-\theta}), \tag{A4}$$

$$1 - F_0(z) = 1 - \exp(-z^{-\theta}). \tag{A5}$$

Direct computation gives

$$\varepsilon(z) = \frac{z \cdot \theta\, z^{-(\theta+1)} \exp(-z^{-\theta})}{1 - \exp(-z^{-\theta})} = \frac{\theta\, z^{-\theta} \exp(-z^{-\theta})}{1 - \exp(-z^{-\theta})}. \tag{A6}$$

Let $u \equiv z^{-\theta}$. Then $\varepsilon = \theta u e^{-u}/(1 - e^{-u})$. Using $1 - e^{-u} = u e^{-u} \sum_{k=0}^{\infty} u^k/(k+1)!$ shows that $\varepsilon = \theta$ exactly when $u \to 0$ (i.e., $z \to \infty$), which is the relevant range for marginal exporters. More precisely, $\varepsilon(z) = \theta + O(z^{-\theta})$, confirming that the elasticity is exactly constant in the Fréchet limit and approximately constant for large z.

A.2 Derivation of the Melitz Trade Elasticity

In the Melitz model with CES preferences (elasticity $\sigma > 1$), a firm with productivity φ in country i earns revenue from sales to country j equal to

$$r_{ij}(\varphi) = A_j \left(\frac{\tau_{ij}\, w_i}{\varphi} \right)^{1-\sigma}, \tag{A7}$$

where A_j is a destination-specific demand shifter. The firm exports to j if and only if variable profits cover the fixed export cost:

$$\frac{r_{ij}(\varphi)}{\sigma} \geq w_i\, f_{ij}. \tag{A8}$$

This defines the zero-profit cutoff

$$\varphi^*_{ij} = \left(\frac{\sigma\, w_i\, f_{ij}}{A_j} \right)^{\frac{1}{\sigma-1}} \tau_{ij}\, w_i, \tag{A9}$$

which is proportional to τ_{ij}. Aggregate bilateral trade from i to j is

$$\begin{aligned} X_{ij} &= M_i \int_{\varphi^*_{ij}}^{\infty} r_{ij}(\varphi)\, g(\varphi)\, d\varphi \\ &= M_i\, A_j\, (\tau_{ij}\, w_i)^{1-\sigma}\, (\varphi^*_{ij})^{\sigma-1}\, \Phi(\varphi^*_{ij}), \end{aligned} \tag{A10}$$

where

$$\Phi(\varphi^*) \equiv \int_{\varphi^*}^{\infty} \left(\frac{\varphi}{\varphi^*} \right)^{\sigma-1} g(\varphi)\, d\varphi \tag{A11}$$

collects the revenue contribution of all firms above the cutoff. Since $\varphi^*_{ij} \propto \tau_{ij}$, the gravity elasticity requires differentiating $\ln X_{ij}$ with respect to $\ln \tau_{ij}$. The key step is the derivative

of Φ. Applying the Leibniz rule:

$$\begin{aligned}\frac{d\Phi}{d\varphi^*} &= -g(\varphi^*)\cdot 1 + \int_{\varphi^*}^{\infty} \frac{\partial}{\partial\varphi^*}\left(\frac{\varphi}{\varphi^*}\right)^{\sigma-1} g(\varphi)\,d\varphi \\ &= -g(\varphi^*) - \frac{\sigma-1}{\varphi^*}\int_{\varphi^*}^{\infty}\left(\frac{\varphi}{\varphi^*}\right)^{\sigma-1} g(\varphi)\,d\varphi \\ &= -g(\varphi^*) - \frac{\sigma-1}{\varphi^*}\,\Phi(\varphi^*). \end{aligned} \tag{A12}$$

Converting to log derivatives:

$$\frac{d\ln\Phi}{d\ln\varphi^*} = \frac{\varphi^*}{\Phi}\cdot\frac{d\Phi}{d\varphi^*} = -\frac{\varphi^*\,g(\varphi^*)}{\Phi(\varphi^*)} - (\sigma-1). \tag{A13}$$

The gravity elasticity is therefore

$$-\frac{\partial\ln X_{ij}}{\partial\ln\tau_{ij}} = (\sigma-1) - \frac{d\ln\Phi}{d\ln\varphi^*} = (\sigma-1) + \frac{\varphi^*\,g(\varphi^*)}{\Phi(\varphi^*)}, \tag{A14}$$

which is Equation (4). Under Pareto, $g(\varphi) = \kappa\,\varphi_{\min}^{\kappa}\,\varphi^{-(\kappa+1)}$, and direct integration gives

$$\Phi(\varphi^*) = \frac{\kappa}{\kappa-(\sigma-1)}\left(\frac{\varphi_{\min}}{\varphi^*}\right)^{\kappa}, \tag{A15}$$

$$\frac{\varphi^*\,g(\varphi^*)}{\Phi(\varphi^*)} = \kappa-(\sigma-1), \tag{A16}$$

so $\varepsilon(\varphi^*) = (\sigma-1) + [\kappa-(\sigma-1)] = \kappa$, which is constant.

A.3 Proof of Proposition 1

Proof. From Equation (2), the absolute error satisfies

$$|\mathcal{E}_{\mathrm{EK}}| = \left|\int_{\ln\lambda}^{\ln\lambda'} \frac{\varepsilon(z)-\theta}{\theta\cdot\varepsilon(z)}\,dz\right| \le \int_{\ln\lambda}^{\ln\lambda'} \frac{|\varepsilon(z)-\theta|}{\theta\cdot\varepsilon(z)}\,dz. \tag{A17}$$

Substituting $\eta(z) = \varepsilon(z)-\theta$, the integrand becomes

$$\frac{|\eta(z)|}{\theta\cdot(\theta+\eta(z))}. \tag{A18}$$

Since $\theta \geq \sigma - 1 \geq 1$ (for $\sigma \geq 2$), we have

$$\frac{|\eta(z)|}{\theta \cdot (\theta + \eta(z))} \leq \frac{|\eta(z)|}{\theta + \eta(z)}. \tag{A19}$$

Taking the supremum over $z \in [z_L, z_H]$ and evaluating the remaining integral:

$$|\mathcal{E}_{\text{EK}}| \leq \sup_{z \in [z_L, z_H]} \frac{|\eta(z)|}{\theta + \eta(z)} \cdot \left| \int_{\ln \lambda}^{\ln \lambda'} dz \right| = \sup_{z \in [z_L, z_H]} \frac{|\eta(z)|}{\theta + \eta(z)} \cdot |\ln \lambda'|$$

A.4 Proof of Proposition 2

Proof. From Equation (2), the absolute error satisfies

$$|\mathcal{E}_{\text{Melitz}}| \leq \int_{\ln \lambda}^{\ln \lambda'} \frac{|\varepsilon(\varphi) - \kappa|}{\kappa \cdot \varepsilon(\varphi)} \, d\varphi. \tag{A20}$$

Substituting the wedge relation from Equation (6):

$$\frac{|\varepsilon - \kappa|}{\kappa \cdot \varepsilon} = \frac{[\kappa - (\sigma - 1)]}{\kappa \cdot \varepsilon} \cdot \frac{|\rho - \psi|}{1 + \psi}. \tag{A21}$$

From Equation (4) and Assumption 2, $\varepsilon(\varphi^*) \geq \sigma - 1 > 0$, so

$$\frac{[\kappa - (\sigma - 1)]}{\kappa \cdot \varepsilon} \leq \frac{\kappa - (\sigma - 1)}{(\sigma - 1)\,\kappa}. \tag{A22}$$

Combining and taking the supremum over $\varphi \in [\varphi_L, \varphi_H]$:

$$|\mathcal{E}_{\text{Melitz}}| \leq \frac{\kappa - (\sigma - 1)}{(\sigma - 1)\,\kappa} \cdot \sup_{\varphi \in [\varphi_L, \varphi_H]} \frac{|\rho(\varphi) - \psi(\varphi)|}{1 + \psi(\varphi)} \cdot |\ln \lambda'|$$

A.5 Proof of Lemma 1

Proof. EK class. Under $F_0 \in 2\text{RV}(-\theta, \beta)$, the survival function satisfies, for $t > 0$:

$$1 - F_0(tz) = \big(1 - F_0(z)\big) \cdot t^{-\theta} \left[1 + A(z) \frac{t^\beta - 1}{\beta} + o\big(A(z)\big) \right]. \tag{A23}$$

Differentiating both sides with respect to t and evaluating at $t = 1$ yields

$$z\, f_0(z) = \big(1 - F_0(z)\big) \cdot \big[\theta + (\theta + \beta - 1)\, A(z) + o(A(z))\big]. \tag{A24}$$

Dividing by $1 - F_0(z)$:

$$\varepsilon(z) = \theta + (\theta + \beta - 1)\, A(z) + o(A(z)). \tag{A25}$$

Hence $\eta(z) = \varepsilon(z) - \theta = O(A(z))$, and since $\theta + \eta(z) \to \theta > 0$:

$$\frac{|\eta(z)|}{\theta + \eta(z)} = O\big(|A(z)|\big) \quad \text{as } z \to \infty. \tag{A26}$$

Melitz class. Under $\bar{G} \in \mathrm{2RV}(-\kappa, \beta)$, differentiating the 2RV representation and comparing with the Pareto density $g_P(\varphi) = \kappa\, \varphi_{\min}^{\kappa}\, \varphi^{-(\kappa+1)}$ yields

$$\rho(\varphi) \equiv \frac{g(\varphi)}{g_P(\varphi)} - 1 = \frac{\kappa + \beta - 1}{\kappa}\, A(\varphi)\big(1 + o(1)\big). \tag{A27}$$

For the revenue-weighted aggregate, substituting the 2RV expansion into $\Phi(\varphi^*)$ and integrating term by term gives

$$\psi(\varphi^*) \equiv \frac{\Phi(\varphi^*)}{\Phi_P(\varphi^*)} - 1 = \frac{\kappa - \sigma + 1 + \beta}{\kappa - \sigma + 1}\, A(\varphi^*)\big(1 + o(1)\big). \tag{A28}$$

The sufficiency wedge is

$$\rho(\varphi^*) - \psi(\varphi^*) = \left[\frac{\kappa + \beta - 1}{\kappa} - \frac{\kappa - \sigma + 1 + \beta}{\kappa - \sigma + 1}\right] A(\varphi^*)\big(1 + o(1)\big) = O\big(A(\varphi^*)\big), \tag{A29}$$

with a leading coefficient that depends on κ, σ, and β. Since $A(\varphi^*) \to 0$ as $\varphi^* \to \infty$, the wedge vanishes at rate $|A(\varphi^*)|$. □

A.6 Proof of Remark 1

Proof. EK class. The "if" direction is immediate: under Fréchet, $\varepsilon(z) = \theta$ for all z, so gravity is log-linear and $\beta_2 = 0$. For the "only if" direction, suppose $\beta_2 = 0$. Then

$\varepsilon(z) = \theta$ for all margins spanned by observed bilateral pairs. From Equation (3):

$$\frac{z\, f_0(z)}{1 - F_0(z)} = \theta \quad \Longrightarrow \quad \frac{d \ln(1 - F_0)}{d \ln z} = -\theta. \tag{A30}$$

This is a first-order linear equation in $\ln(1 - F_0)$ with unique solution

$$1 - F_0(z) = C\, z^{-\theta} \tag{A31}$$

for some constant $C > 0$, which is the Fréchet survival function.

Melitz class. The "if" direction is again immediate. For the "only if" direction, $\beta_2 = 0$ implies $\varepsilon(\varphi^*) = \kappa$ for all observed cutoffs. From Equation (4):

$$\frac{\varphi^*\, g(\varphi^*)}{\Phi(\varphi^*)} = \kappa - (\sigma - 1) \quad \Longrightarrow \quad \frac{d \ln \Phi}{d \ln \varphi^*} = -\kappa. \tag{A32}$$

The unique solution is $\Phi(\varphi^*) = C\, (\varphi^*)^{-\kappa}$, which implies $\bar{G}(\varphi) \propto \varphi^{-\kappa}$. □

B Additional Tables

Table B1 reports aggregate bounds for both model classes across $\sigma \in \{3, 4, 5\}$. Both bounds scale as $1/(\sigma - 1)$ through the denominators of Equations (16) and (17). The qualitative conclusion is unchanged across all specifications.

Table B1: Aggregate bounds by σ

	EK bound (pp)			Melitz bound (pp)		
Year	$\sigma = 3$	$\sigma = 4$	$\sigma = 5$	$\sigma = 3$	$\sigma = 4$	$\sigma = 5$
2015	0.09	0.06	0.04	0.11	0.07	0.05
2016	0.10	0.06	0.05	0.12	0.08	0.06
2017	0.06	0.04	0.03	0.07	0.05	0.04
2018	0.04	0.03	0.02	0.05	0.04	0.03
2019	0.08	0.06	0.04	0.11	0.07	0.05

Notes: EK bound from Equation (16); Melitz bound from Equation (17). Both evaluated at $|\Delta \ln \hat{\lambda}| = 0.013$. Both bounds scale as $1/(\sigma - 1)$.

Table B2 reports the gravity diagnostic for all HS2 sectors with at least 100 bilateral pairs, sorted by p-value. Sectors with $p < 0.05$ are indicated by †.

Table B2: Full sectoral gravity diagnostic (HS2, 2019). †: $p < 0.05$. Bounds evaluated at $\sigma = 4$ and $|\Delta \ln \hat{\lambda}| = 0.013$. "—" indicates that the Melitz bound is undefined because the implied pooled elasticity from the quadratic specification is non-positive.

						Bound (pp)	
HS2	Sector	N	$\hat{\varepsilon}_{\text{pool}}$	$\hat{\beta}_2$	p	EK	Melitz
07†	Edible vegetables	744	1.453	−0.251	<0.001	0.65	—
03†	Fish	704	0.726	0.301	<0.001	0.78	0.14
12†	Oil seeds	776	0.866	0.440	<0.001	1.13	0.15
93†	Arms/ammunition	662	0.320	0.504	<0.001	1.30	0.16
42†	Leather articles	796	0.307	0.222	<0.001	0.57	0.15
80†	Tin	625	0.857	−0.289	<0.001	0.75	—
95†	Toys/games	792	0.841	0.211	<0.001	0.54	0.13
23†	Food residues	730	0.976	0.215	<0.001	0.55	0.13
15†	Fats/oils	790	1.034	0.232	<0.001	0.60	0.13
67†	Feathers/down	692	0.879	0.449	<0.001	1.16	0.15
11†	Milling products	756	1.319	−0.176	<0.001	0.45	—

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						Bound (pp)	
HS2	Sector	N	$\hat{\varepsilon}_{\text{pool}}$	$\hat{\beta}_2$	p	EK	Melitz
09†	Coffee/tea/spices	787	0.915	0.174	<0.001	0.45	0.12
26†	Ores	699	1.408	−0.182	<0.001	0.47	—
46†	Straw manufactures	637	0.825	0.321	<0.001	0.83	0.14
52†	Cotton	742	0.750	0.226	0.001	0.58	0.13
06†	Live plants	652	1.340	−0.144	0.001	0.37	—
79†	Zinc	715	1.284	−0.176	0.002	0.45	—
04†	Dairy	713	1.459	−0.114	0.003	0.29	—
44†	Wood	801	0.958	0.177	0.003	0.46	0.12
10†	Cereals	694	1.280	0.210	0.003	0.54	0.12
08†	Edible fruit	752	1.537	−0.190	0.003	0.49	—
49†	Printed books	801	1.004	0.209	0.003	0.54	0.13
66†	Umbrellas	659	1.172	0.171	0.003	0.44	0.12
33†	Perfumery	799	0.864	−0.136	0.004	0.35	—
75†	Nickel	726	0.475	−0.242	0.006	0.62	—
05†	Other animal prod.	671	0.693	0.147	0.006	0.38	0.13
16†	Meat/fish prep.	714	1.152	0.157	0.006	0.41	0.11
64†	Footwear	793	0.870	0.161	0.007	0.42	0.12
18†	Cocoa	765	1.142	−0.121	0.010	0.31	—
68†	Stone/cement art.	785	1.023	0.101	0.014	0.26	0.10
56†	Wadding/felt	759	1.012	−0.092	0.018	0.24	—
13†	Lac/gums/resins	714	0.524	0.145	0.018	0.37	0.13
74†	Copper	793	1.210	−0.141	0.019	0.36	—
62†	Woven apparel	801	0.845	0.132	0.021	0.34	0.12
72†	Iron/steel	784	1.160	−0.083	0.023	0.21	—
17†	Sugars	773	1.397	−0.123	0.032	0.32	—
97†	Art/antiques	768	0.156	−0.212	0.037	0.55	—
92†	Musical instruments	750	0.491	0.109	0.040	0.28	0.13
59†	Coated fabrics	758	0.958	−0.084	0.047	0.22	—
82	Tools	806	0.710	0.073	0.051	0.19	0.10
53	Vegetable fibers	640	0.419	0.160	0.055	0.41	0.14
51	Wool	672	0.838	−0.199	0.057	0.51	—
71	Precious metals	793	0.342	−0.238	0.062	0.61	—
31	Fertilizers	703	1.050	0.080	0.071	0.21	0.09
86	Railway vehicles	741	1.190	−0.121	0.078	0.31	—
36	Explosives	604	0.572	0.110	0.084	0.28	0.12

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						Bound (pp)	
HS2	Sector	N	$\hat{\varepsilon}_{\text{pool}}$	$\hat{\beta}_2$	p	EK	Melitz
34	Soap/wax	798	1.104	−0.050	0.086	0.13	0.39
22	Beverages	793	0.804	−0.081	0.099	0.21	—
63	Other textiles	802	1.104	0.099	0.104	0.25	0.10
90	Optical/medical	806	0.547	−0.055	0.107	0.14	—
39	Plastics	807	1.011	−0.049	0.128	0.13	0.52
47	Paper pulp	643	0.979	−0.111	0.138	0.29	—
01	Live animals	667	1.699	−0.115	0.150	0.30	—
58	Special woven fabrics	745	0.937	−0.097	0.152	0.25	—
91	Clocks/watches	777	0.321	0.122	0.161	0.31	0.14
57	Carpets	745	1.104	0.066	0.162	0.17	0.08
54	Man-made filaments	756	0.840	−0.073	0.163	0.19	—
19	Cereal preparations	788	1.216	−0.060	0.177	0.16	0.52
84	Machinery	808	0.714	−0.040	0.203	0.10	1.17
35	Albuminoidal subst.	773	0.897	−0.037	0.227	0.09	0.29
29	Organic chemicals	796	0.754	−0.043	0.240	0.11	1.25
14	Vegetable plaiting	585	0.769	−0.100	0.250	0.26	—
27	Mineral fuels	791	1.269	−0.068	0.262	0.18	0.81
81	Other base metals	729	0.494	−0.046	0.309	0.12	—
65	Headgear	783	0.816	0.059	0.342	0.15	0.09
61	Knitted apparel	801	1.053	0.054	0.373	0.14	0.07
28	Inorganic chemicals	787	0.845	−0.036	0.386	0.09	0.33
32	Tanning/dyeing	788	0.951	−0.024	0.395	0.06	0.11
94	Furniture	794	1.082	0.043	0.406	0.11	0.06
24	Tobacco	677	0.899	0.096	0.413	0.25	0.10
78	Lead	638	1.218	−0.065	0.439	0.17	0.81
73	Iron/steel articles	804	0.991	0.031	0.454	0.08	0.05
02	Meat	580	1.235	0.089	0.456	0.23	0.09
60	Knitted fabrics	707	1.136	−0.058	0.507	0.15	0.66
76	Aluminum	801	1.020	0.027	0.521	0.07	0.05
21	Misc food prep.	799	1.025	0.025	0.543	0.06	0.05
55	Man-made staple	741	0.676	0.037	0.546	0.09	0.08
25	Salt/stone/cement	788	1.243	0.024	0.553	0.06	0.04
87	Vehicles	798	0.913	−0.029	0.575	0.08	0.16
41	Hides/skins	735	0.918	−0.033	0.584	0.09	0.22
96	Misc manufactured	793	0.832	0.024	0.628	0.06	0.05

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						Bound (pp)	
HS2	Sector	N	$\hat{\varepsilon}_{\text{pool}}$	$\hat{\beta}_2$	p	EK	Melitz
69	Ceramics	782	0.887	0.020	0.640	0.05	0.04
88	Aircraft	771	0.373	0.031	0.655	0.08	0.09
38	Chemical products	793	0.849	−0.011	0.659	0.03	0.04
45	Cork	625	1.301	0.039	0.670	0.10	0.05
48	Paper	801	1.243	−0.012	0.730	0.03	0.03
30	Pharmaceuticals	795	0.350	0.017	0.751	0.04	0.07
50	Silk	567	0.720	0.039	0.771	0.10	0.07
20	Vegetable prep.	778	0.995	−0.012	0.781	0.03	0.04
43	Furskins	664	0.501	−0.037	0.794	0.10	—
89	Ships	705	0.843	−0.023	0.818	0.06	0.12
37	Photographic goods	663	0.638	0.013	0.867	0.03	0.04
40	Rubber	804	0.857	−0.005	0.913	0.01	0.02
83	Misc base metal art.	792	0.980	−0.004	0.925	0.01	0.01
85	Electrical machinery	810	0.826	−0.001	0.979	<0.01	<0.01
70	Glass	795	0.994	−0.000	0.998	<0.01	<0.01